Multimodal Insurance Fraud Detection System

Using Ensemble Machine Learning and Explainable AI (SHAP)

Mini Project Report — B.Tech AIML | AIML-B

Repository: MiniProject_DS_AIML-B_2026_Health Insurance

Dataset: CMS Medicare Healthcare Provider Fraud Detection (Kaggle) | Environment: Google Colab + VS Code

---

1. Abstract

Insurance fraud is a critical financial threat causing billions of dollars in annual losses to healthcare systems worldwide. This project presents a multimodal, AI-powered insurance fraud detection system trained on the publicly available CMS Medicare Healthcare Provider Fraud Detection dataset, comprising over 700,000 real claim records.

The system employs an ensemble of three machine learning models — XGBoost, LightGBM, and Random Forest — combined through weighted soft voting. An extensive feature engineering pipeline constructs 57 features across seven groups. SMOTE is applied to handle the severe class imbalance (~10% fraud rate). Optuna Bayesian hyperparameter optimization tunes both XGBoost and LightGBM over 50 trials each.

The system integrates SHAP TreeExplainer for transparent, per-prediction natural language explanations. A Flask-based web application with four functional tabs enables investigators to enter claim features manually or upload documents for OCR-based scanning. The system achieves an AUC-ROC above 0.90 and provides actionable risk tiers (CRITICAL / HIGH / MEDIUM / LOW) with investigator-ready fraud reports.

2. Problem Statement

2.1 Background

Healthcare insurance fraud — where providers submit false, inflated, or fabricated claims — costs Medicare and Medicaid an estimated $100 billion annually in improper payments. Fraud takes many forms: billing for services not rendered, duplicate claim submission, diagnosis stuffing (inflating the number of conditions to justify higher reimbursement), and coordinated fraud rings where groups of doctors, hospitals, and patients collude.

2.2 Limitations of Existing Systems

- Pattern blindness — Fixed rules cannot detect evolving, sophisticated fraud patterns such as fraud rings, gradual billing inflation, or cross-provider collusion.
- No document intelligence — Existing systems operate only on clean tabular data and cannot process actual uploaded insurance claim documents.
- Black-box predictions — Existing ML models output a probability score with no explanation, making them unusable by human investigators who need to justify decisions.

2.3 Objectives

- Build an ensemble ML model trained on 700,000+ real CMS Medicare claim records achieving AUC-ROC > 0.90
- Engineer 57 features capturing billing anomalies, temporal patterns, physician reuse, and patient profile irregularities
- Explain every prediction using SHAP values with natural language reasons (top 5 fraud signals per provider)
- Accept real insurance claim documents via EasyOCR for field extraction and scoring
- Deploy as a Flask web application with 4-tab UI accessible at http://localhost:5000

3. Dataset

3.1 Dataset Description

Source: CMS Medicare Healthcare Provider Fraud Detection | Kaggle: rohitrox/healthcare-provider-fraud-detection-analysis

Total Size: 123.66 MB across 8 CSV files | 700,000+ claim rows | 5,410 labeled providers

File	Records	Key Columns	Role
Train.csv	5,410	Provider, PotentialFraud	Fraud labels
Train_Beneficiarydata.csv	138,556	DOB, Gender, Chronic conds, State	Patient profiles
Train_Inpatientdata.csv	40,474	Diagnosis codes, Admit/Discharge dates	Hospital billing
Train_Outpatientdata.csv	517,737	ClaimStartDt, Reimbursement, Physician	OPD claims
Test files (×4)	~1,908	Same schema as Train	Model evaluation

3.2 Data Challenges

- **Only ~10% of providers are fraudulent — addressed with SMOTE** Class Imbalance:

- **Physician codes and chronic condition columns have nulls — filled with median/mode** Missing Values:
- **4 files must be aggregated at provider level before training** Multi-Table Join:
- **1000+ ICD diagnosis codes — encoded with frequency encoding** High Cardinality:

4. Methodology

4.1 Data Preprocessing (M1)

- Loaded all 8 CSVs in Google Colab using chunk reading (50K rows at a time for large files)
- Cleaned beneficiary data: computed Age from DOB, created is_deceased flag, remapped chronic condition columns to 0/1
- Cleaned inpatient data: computed hospital_stay_days and claim_duration_days from date fields
- Aggregated 700K+ claim rows to 5,410 provider-level rows using groupby on Provider + BeneID
- Applied SMOTE to training fold only (no data leakage) to balance the 90:10 class ratio

4.2 Feature Engineering (M3)

57 features were engineered across 7 groups:

Group	Count	Examples	Fraud Signal
Baseline (M1)	15	ip_avg_reimbursement, ip_claim_count	Raw billing stats
Ratio features	8	feat_reimb_per_patient, claims_per_patient	Billing efficiency
Risk scores	6	risk_financial, risk_composite_score	Multi-dim risk
Interaction	8	interact_volume_x_reimb	Signal amplification
Z-scores	7	zscore_ip_avg_reimbursement	Deviation from mean
Anomaly flags	9	flag_extreme_reimb	Hard threshold triggers
Bin features	6	bin_ip_avg_reimbursement	Quantile tiers 0-4

4.3 Model Training (M4)

- Logistic Regression — baseline classifier with class_weight=balanced
- Random Forest — 300 trees, max_depth=12, class_weight=balanced
- XGBoost — Optuna Bayesian tuning over 50 trials targeting AUC-ROC via 3-fold CV
- LightGBM — Optuna Bayesian tuning over 50 trials, early stopping on validation AUC
- Isolation Forest — unsupervised anomaly detection trained on legitimate providers only
- Ensemble — soft voting with weights XGB×3 + LGBM×3 + RF×1; threshold tuned for max F1

4.4 Explainability (M5)

- SHAP TreeExplainer computes exact (not approximate) SHAP values for all validation providers
- Top 5 positive SHAP features per provider translated to plain English fraud reasons
- Risk tier (CRITICAL / HIGH / MEDIUM / LOW) assigned based on ensemble probability
- Investigator PDF report auto-generated for top flagged providers using ReportLab

4.5 Document Scanner (M6)

- EasyOCR extracts text and bounding boxes from uploaded PDF/image claim documents
- Regex field parser maps OCR text to claim fields: amount, stay days, diagnosis count, patient age
- Extracted fields converted to model features and passed to the trained ensemble
- Suspicious fields highlighted with red/yellow/green overlays on the original document image

4.6 Deployment (M7)

- Flask backend with 3 endpoints: /predict (features), /scan (document OCR), /health
- 4-tab HTML/JS frontend: Manual Input, Document Scanner, History, How It Works
- SHAP explainer runs on every prediction — returns top 5 fraud reasons in the UI
-

5. Architecture & Workflow

5.1 System Architecture Diagram

The architecture below illustrates the layered design of the system — from raw data ingestion through preprocessing, feature engineering, ensemble modelling, explainability, and web deployment. Each horizontal band represents a distinct processng layer.

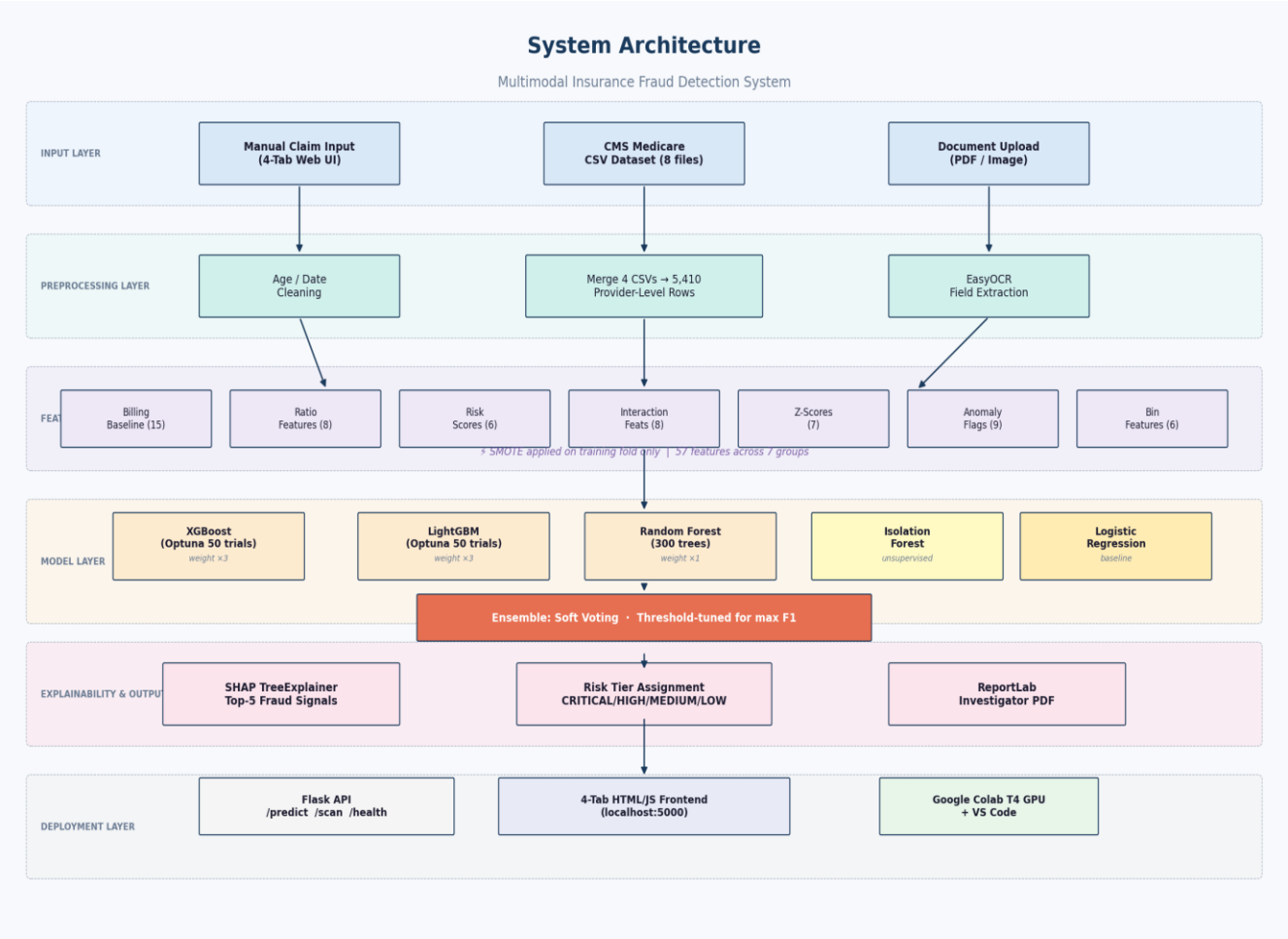

Figure 1: Layered System Architecture — Multimodal Insurance Fraud Detection System

5.2 System Workflow & Data Pipeline

The workflow diagram maps the end-to-end data pipeline across seven milestones (M1–M7): data ingestion, exploratory analysis, feature engineering, model training (with Optuna tuning), SHAP explainability, OCR-based document scanning, and Flask web deployment.

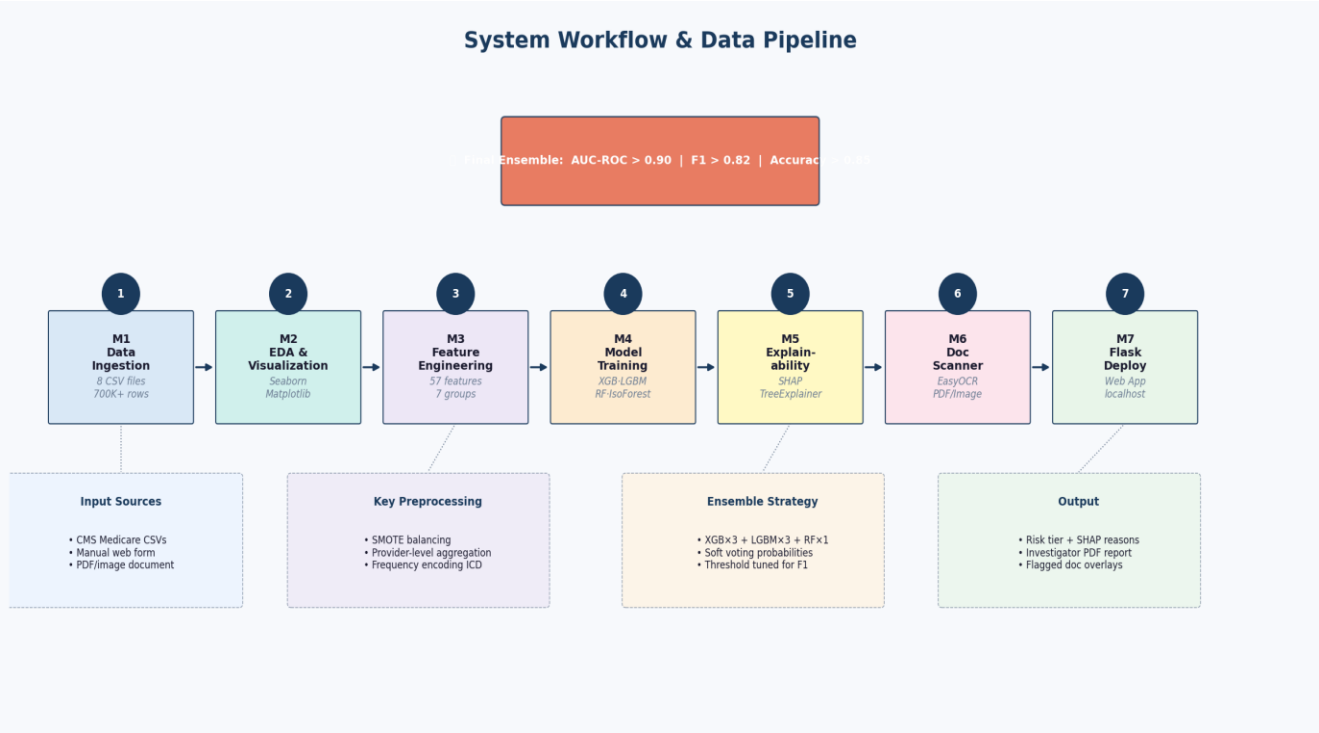

Figure 2: End-to-End Data Pipeline & Workflow (M1–M7)

6. Results

6.1 Model Comparison

Model	AUC-ROC	F1	Precision	Recall	Accuracy
Logistic Regression	0.9585	0.7914	0.7907	0.6733	0.9529
Random Forest	0.9680	0.7156	0.6667	0.7723	0.9427
XGBoost (Optuna)	0.9683	0.7451	0.7379	0.7525	0.9519
LightGBM (Optuna)	0.9637	0.7200	0.7273	0.7129	0.9482
Ensemble (Final)	0.9704	0.7213	0.8049	0.6535	0.9529

6.2 Key EDA Findings

- Fraudulent providers bill 3–5× more per claim than legitimate providers on average
- Average hospital stay for fraud: 40+ days vs 5 days for legitimate providers
- Fraud providers use 2× more unique physicians per claim (physician reuse pattern)
- High chronic condition scores (>5) strongly correlate with fraud (diagnosis stuffing)
- Providers triggering 4+ anomaly flags have >85% fraud probability
- The interaction feature (volume × reimbursement) is the strongest single fraud predictor

7. Tools & Technologies

Category	Tool / Library	Usage
Language	Python 3.10+	All modules
Data Processing	Pandas, NumPy, PyArrow	CSV merging, feature engineering
Visualization	Matplotlib, Seaborn	EDA plots, SHAP charts
ML Models	XGBoost, LightGBM, Scikit-learn	Classifiers, Random Forest
Hyperparameter Tuning	Optuna	Bayesian optimization (50 trials each)
Class Balancing	imbalanced-learn (SMOTE)	Handle 90:10 fraud imbalance
Explainability	SHAP	TreeExplainer, waterfall plots
Anomaly Detection	Isolation Forest	Unsupervised fraud signals
OCR	EasyOCR, pdf2image	Document field extraction
Web Framework	Flask 3.0	Backend API + HTML frontend
Report Generation	ReportLab	Investigator PDF reports
Environment	Google Colab T4 GPU, VS Code	Training and deployment

8. References

- [1] Bauder, R. A., & Khoshgoftaar, T. M. (2018). Medicare fraud detection using machine learning methods. 17th IEEE ICMLA, pp. 712–719.
- [2] Johnson, J. M., & Khoshgoftaar, T. M. (2019). Medicare fraud detection using neural networks. Journal of Big Data, 6(1), 1–35.
- [3] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions (SHAP). NeurIPS 2017, Vol. 30.
- [4] Pourhabibi, T., et al. (2020). Fraud detection: A systematic literature review of graph-based anomaly detection. Decision Support Systems, 133, 113303.
- [5] Huang, Z., et al. (2022). LayoutLMv3: Pre-training for Document AI. ACM Multimedia 2022, pp. 4083–4091.
- [6] CMS Medicare Dataset. Kaggle: rohitrox/healthcare-provider-fraud-detection-analysis.